

ANSIBLE-TASK-3

TASKS:

- ❖ Write a single ansible playbook which will install apache and nginx.
Note: Playbook should not be hardcoded and pass the variables from different file.
- ❖ Ansible playbook to create 10 different directories with minimal code and directory names should be passed as variables.
- ❖ Ansible playbook to copy ssh-keygen from master to worker nodes.
Note: a) Provision new 3 ec2 machines, one master and two worker nodes. B) Create common user called ansadm and provide sudo privileges on 3 ec2 instances. c) Create ssh-keygen in master and your playbook should copy the keygen making it password less authentication.
- ❖ Ansible playbook to inject ansible vault variables.

1. Write a single ansible playbook which will install apache and nginx.
Note: Playbook should not be hardcoded and pass the variables from different file.

```
ubuntu@ip-172-31-24-53:~/playbooks$ cd ..
ubuntu@ip-172-31-24-53:~$ ansible all --ping
[WARNING]: Invalid characters were found in group names but not replaced, use --vvvv to see details
172.31.20.17 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
172.31.31.214 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
ubuntu@ip-172-31-24-53:~$
```

Webserver.yml

- name: Install and configure Apache & Nginx

hosts: all

become: yes

vars_files:

- vars.yml

tasks:

Update & Install packages

- name: Update apt cache

apt:

update_cache: yes

- name: Install Apache

apt:

name: apache2

state: present

- name: Install Nginx

apt:

name: nginx

state: present

Configure Apache

- name: Set Apache listen port

lineinfile:

path: /etc/apache2/ports.conf

regexp: '^Listen'

line: "Listen {{ apache_port }}"

notify: Restart Apache

- name: Configure Apache vhost

copy:

dest: /etc/apache2/sites-enabled/000-default.conf

content: |

```
<VirtualHost *:{{ apache_port }}>
    DocumentRoot /var/www/html
    <Directory /var/www/html>
        AllowOverride All
        Require all granted
    </Directory>
</VirtualHost>
```

notify: Restart Apache

Configure Nginx (Ubuntu safe)

- name: Disable default nginx site (port 80)

file:

path: /etc/nginx/sites-enabled/default

state: absent

notify: Restart Nginx

- name: Configure Nginx on port 81

copy:

dest: /etc/nginx/conf.d/nginx81.conf

content: |

```
server {  
    listen {{ nginx_port }};  
    server_name _;  
    root /usr/share/nginx/html;  
    index index.html;  
}
```

notify: Restart Nginx

Start and enable services

- name: Start and enable Apache

systemd:

name: apache2

state: started

enabled: yes

- name: Start and enable Nginx

systemd:

name: nginx

state: started

enabled: yes

handlers:

- name: Restart Apache

service:

name: apache2

state: restarted

- name: Restart Nginx

service:

name: nginx

state: restarted

```
ubuntu@ip-172-31-24-53: ~/playbook
--
- name: Install and configure Apache & Nginx
  hosts: all
  become: yes
  vars_files:
    - vars.yml

  tasks:
    # -----
    # Update & Install packages
    # -----
    - name: Update apt cache
      apt:
        update_cache: yes

    - name: Install Apache
      apt:
        name: apache2
        state: present

    - name: Install Nginx
      apt:
        name: nginx
        state: present

    # -----
    # Configure Apache
    # -----
    - name: Set Apache listen port
      lineinfile:
        path: /etc/apache2/ports.conf
        regexp: '^#listen'
        line: "listen {{ apache_port }}"
        notify: Restart Apache

    - name: Configure Apache vhost
      copy:
        dest: /etc/apache2/sites-enabled/000-default.conf
        content: |
          <VirtualHost *:{{ apache_port }}>
            DocumentRoot /var/www/html
            <Directory /var/www/html>
              AllowOverride All
              Require all granted
            </Directory>
          </VirtualHost>
        notify: Restart Apache

    # -----
    # Configure Nginx (ubuntu safe)
    # -----
    - name: Disable default nginx site (port 80)
      file:
        path: /etc/nginx/sites-enabled/default
        state: absent
        notify: Restart Nginx
```

Vars.yml

apache_port: 80

nginx_port: 81

```
ubuntu@ip-172-31-24-53: ~/playbook
--
apache_port: 80
nginx_port: 81
--
```

```
ubuntu@ip-172-31-24-53:~/playbook$ ansible-playbook appserver.yml
[WARNING]: Invalid characters were found in group names but not replaced, use -vvvv to see details

PLAY [Install and configure Apache & Nginx] *****

TASK [Gathering Facts] *****
ok: [172.31.20.77]
ok: [172.31.31.214]

TASK [Update apt cache] *****
changed: [172.31.20.77]
changed: [172.31.31.214]

TASK [Install Apache] *****
changed: [172.31.31.214]
changed: [172.31.20.77]

TASK [Install Nginx] *****
changed: [172.31.31.214]
changed: [172.31.20.77]

TASK [Set Apache listen port] *****
ok: [172.31.31.214]
ok: [172.31.20.77]

TASK [Configure Apache vhost] *****
changed: [172.31.31.214]
changed: [172.31.20.77]

TASK [Disable default nginx site (port 80)] *****
changed: [172.31.31.214]
changed: [172.31.20.77]

TASK [Configure Nginx on port 81] *****
changed: [172.31.31.214]
changed: [172.31.20.77]

TASK [Start and enable Apache] *****
ok: [172.31.20.77]
ok: [172.31.31.214]

TASK [Start and enable Nginx] *****
changed: [172.31.31.214]
changed: [172.31.20.77]

RUNNING HANDLER [Restart Apache] *****
changed: [172.31.31.214]
changed: [172.31.20.77]

RUNNING HANDLER [Restart Nginx] *****
changed: [172.31.31.214]
changed: [172.31.20.77]

PLAY RECAP *****
172.31.20.77 : ok=12 changed=9 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
172.31.31.214 : ok=12 changed=9 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0

ubuntu@ip-172-31-24-53:~/playbook$ vi |
```

2. Ansible playbook to create 10 different directories with minimal code and directory names should be passed as variables.

- name: Create multiple directories using variables

hosts: all

become: true

vars:

dir_list: #task2

- /opt/app1

- /opt/app2

- /opt/app3

- /opt/app4

- /opt/app5

- /opt/app6

- /opt/app7

- /opt/app8

- /opt/app10

- name: Create directories

```
path: "{{ item }}"
```

state: directory

mode: '0755'

```
loop: "{{ dir_list }}"
```

```
ubuntu@ip-172-31-24-53: ~/playbook
--
- name: Create multiple directories using variables
  hosts: all
  become: true

  vars:
    dir_list:      #task2
    - /opt/app1
    - /opt/app2
    - /opt/app3
    - /opt/app4
    - /opt/app5
    - /opt/app6
    - /opt/app7
    - /opt/app8
    - /opt/app9
    - /opt/app10

  tasks:
    - name: Create directories
      file:
        path: "[[ item ]]"
        state: directory
        mode: '0755'
        loop: "[[ dir_list ]]"
```

```

[WARNING]: Invalid characters were found in group name but not replaced, use -vvvv to see details

TASK [Create multiple directories using variables] *****
TASK [Gathering Facts] *****
172.31.31.214:
ok: [172.31.31.214]
ok: [172.31.20.77]

TASK [Create directories] *****
changed: [172.31.31.214] => (item=/opt/app1)
changed: [172.31.20.77] => (item=/opt/app1)
changed: [172.31.31.214] => (item=/opt/app2)
changed: [172.31.20.77] => (item=/opt/app2)
changed: [172.31.31.214] => (item=/opt/app3)
changed: [172.31.20.77] => (item=/opt/app3)
changed: [172.31.31.214] => (item=/opt/app4)
changed: [172.31.20.77] => (item=/opt/app4)
changed: [172.31.31.214] => (item=/opt/app5)
changed: [172.31.20.77] => (item=/opt/app5)
changed: [172.31.31.214] => (item=/opt/app6)
changed: [172.31.20.77] => (item=/opt/app6)
changed: [172.31.31.214] => (item=/opt/app7)
changed: [172.31.20.77] => (item=/opt/app7)
changed: [172.31.31.214] => (item=/opt/app8)
changed: [172.31.20.77] => (item=/opt/app8)
changed: [172.31.31.214] => (item=/opt/app9)
changed: [172.31.20.77] => (item=/opt/app9)
changed: [172.31.31.214] => (item=/opt/app10)
changed: [172.31.20.77] => (item=/opt/app10)

PLAY RECAP *****
172.31.31.214:
  ok=2 changed=1 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
172.31.20.77:
  ok=2 changed=1 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0

```

```
ubuntu@ip-172-31-31-214:/home$ cd /opt
ubuntu@ip-172-31-31-214:/opt$ ls
app1  app10  app2  app3  app4  app5  app6  app7  app8  app9  tomcat
```

i-031d1cfec380f0060 (worker-1)

PublicIPs: 54.205.241.197 PrivateIPs: 172.31.31.214

```
ubuntu@ip-172-31-20-77:/opt$ ls
app1 app10 app2 app3 app4 app5 app6 app7 app8 app9 tomcat
```

i-09f70fd93f2338ed4 (worker-2)

PublicIPs: 100.53.178.253 PrivateIPs: 172.31.20.77

3. Ansible playbook to copy ssh-keygen from master to worker nodes.

Note: a)Provision new 3 ec2 machines, one master and two worker nodes. b)Create common user called ansadm and provide sudo privileges on 3 ec2 instances. c)Create ssh-keygen in master and your playbook should copy the keygen making it password less authentication.

- name: Create EC2 instances

hosts: localhost

connection: local

gather_facts: no

vars:

region: us-east-1

ami_id: ami-0532be01f26a3de55

instance_type: t2.micro

key_name: arj

security_group: default

tasks:

- name: Create EC2 instances

amazon.aws.ec2_instance:

name: "{{ item.name }}"

key_name: "{{ key_name }}"

instance_type: "{{ instance_type }}"

image_id: "{{ ami_id }}"

region: "{{ region }}"

wait: yes

security_group: "{{ security_group }}"

tags:

Name: "{{ item.name }}"

loop:

- { name: "ansible-master" }

- { name: "worker-1" }

- { name: "worker-2" }

```
name: Create EC2 instances
hosts: localhost
connection: local
gather_facts: no

vars:
  region: us-east-1
  ami_id: ami-0532be01f26a3de55
  instance_type: t2.micro
  key_name: arj
  security_group: default

tasks:
  - name: Create EC2 instances
    amazon.aws.ec2.Instance:
      name: "{{ item.name }}"
      key_name: "{{ key_name }}"
      instance_type: "{{ instance_type }}"
      image_id: "{{ ami_id }}"
      region: "{{ region }}"
      wait: yes
      security_group: "{{ security_group }}"
      tags:
        Name: "{{ item.name }}"
    loop:
      - { name: "ansible-master" }
      - { name: "worker-1" }
      - { name: "worker-2" }
```

```
Last login: Fri Jan 30 10:42:38 2026 from 14.192.14.53
ubuntu@ip-172-31-24-53:~$ ls
aws_awscli_v2.zip  playbook
ubuntu@ip-172-31-24-53:~$ cd playbook/
ubuntu@ip-172-31-24-53:~/playbook$ vi multi_instance.yml
ubuntu@ip-172-31-24-53:~/playbook$ ansible-playbook multi_instance.yml --syntax-check
[WARNING]: Invalid characters were found in group names but not replaced, use -vvvv to see details

playbook: multi_instance.yml
ubuntu@ip-172-31-24-53:~/playbook$ ansible-playbook multi_instance.yml --check
[WARNING]: Invalid characters were found in group names but not replaced, use -vvvv to see details

PLAY [Create EC2 instances] *****
TASK [Create EC2 instances] *****
changed: [localhost] => (item={'name': 'ansible-master'})
changed: [localhost] => (item={'name': 'worker-1'})
changed: [localhost] => (item={'name': 'worker-2'})

PLAY RECAP *****
localhost                : ok=1  changed=1  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0

ubuntu@ip-172-31-24-53:~/playbook$ ansible-playbook multi_instance.yml
[WARNING]: Invalid characters were found in group names but not replaced, use -vvvv to see details

PLAY [Create EC2 instances] *****
TASK [Create EC2 instances] *****
changed: [localhost] => (item={'name': 'ansible-master'})
changed: [localhost] => (item={'name': 'worker-1'})
changed: [localhost] => (item={'name': 'worker-2'})

PLAY RECAP *****
localhost                : ok=1  changed=1  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0

ubuntu@ip-172-31-24-53:~/playbook$ vi multi_instance.yml
ubuntu@ip-172-31-24-53:~/playbook$
```

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager [New](#)

Images

☐	Master_jenkin	i-04e9dd922f531d1cf	Stopped	t2.large	-	View alarms	us-east
☐	MyAnsibleInstance	i-005bfe901b6d46efa	Stopped	t2.micro	-	View alarms	us-east
☐	ansible-master	i-016f1e3b5d92425c4	Running	t2.micro	2/2 checks passed	View alarms	us-east
☒	worker-1	i-0573fb6e2eb008b7e	Running	t2.micro	2/2 checks passed	View alarms	us-east
☒	worker-2	i-09a0572904ab81b8c	Running	t2.micro	2/2 checks passed	View alarms	us-east

3 instances selected

Monitoring

[Configure CloudWatch agent](#)

→ Now install ansible

```
ssh -i "arj.pem" ec2-user@ec2-3-237-101-224.compute-1.amazonaws.com
The authenticity of host 'ec2-3-237-101-224.compute-1.amazonaws.com (3.237.101.224)' can't be established.
eD25519 key fingerprint is SHA256:tscseA0Ioyv00XLoOvrDNKffswEPnID9xPVurPAwJ.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-237-101-224.compute-1.amazonaws.com' (eD25519) to the list of known hosts.
```



```
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-13-122 ~]$ sudo su -
[root@ip-172-31-13-122 ~]# python --version
--bash: python: command not found
[root@ip-172-31-13-122 ~]# python3 --version
python 3.9.25
[root@ip-172-31-13-122 ~]# sudo update -y
sudo: update: command not found
[root@ip-172-31-13-122 ~]# sudo yum update -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-13-122 ~]# yum install ansible -y
Get metadata expiration check: 0:00:16 ago on Fri Jan 30 13:45:43 2026.
No match for argument: ansible
Error: Unable to find a match: ansible
[root@ip-172-31-13-122 ~]# yum install ansible -y
```

249 kb/s | 30 kB 00:00

→ Create one directory ansible and inside that directory create one file called hosts and add worker ips and create groups as well.

```
root@ip-172-31-13-122:~/ansible
[all]
98.92.134.20
34.234.211.44
[worker-1]
98.92.134.20
[worker-2]
34.234.211.44
```

```
[root@ip-172-31-13-122 playbooks]# cd
[root@ip-172-31-13-122 ~]# cat ansible/hosts
[controller]
localhost ansible_connection=local

[workers]
worker-1 ansible_host=172.31.7.125
worker-2 ansible_host=172.31.15.0

[all:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/.ssh/arj.pem
ansible_ssh_common_args='-o StrictHostKeyChecking=no'

[root@ip-172-31-13-122 ~]# |
```

→from downloads we need to send the pem key ec2-user "scp -l arj.pem ec2-user@Master-ip:/home/ec2user"

```

C:\Users\arj\jumbo MINGW64 ~/Downloads (master)
$ ls arj.pem
arj.pem

C:\Users\arj\jumbo MINGW64 ~/Downloads (master)
$ chmod 400 arj.pem

C:\Users\arj\jumbo MINGW64 ~/Downloads (master)
$ scp -i arj.pem arj.pem ec2-user@3.237.101.224:/home/ec2-user
The authenticity of host '3.237.101.224 (3.237.101.224)' can't be established.
ED25519 key fingerprint is SHA256:tcseA010yvdDQXLoOvrvDNvKfSwPEN10xpVRUPAWq.
This host key is known by the following other names/addresses:
  -/.ssh/known_hosts:61: ec2-3-237-101-224.compute-1.amazonaws.com
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.237.101.224' (ED25519) to the list of known hosts.
arj.pem
100% 1678  5.3kB/s  00:0

```

#####

PLAY 1: Create ansadm user on all servers

#####

- name: Create common user and sudo

hosts: all

become: yes

tasks:

- name: Create ansadm user

user:

name: ansadm

shell: /bin/bash

create_home: yes

- name: Allow passwordless sudo

copy:

dest: /etc/sudoers.d/ansadm

content: "ansadm ALL=(ALL) NOPASSWD:ALL"

mode: '0440'

#####

PLAY 2: Generate SSH key on controller (localhost)

#####

- name: Generate SSH key on controller

hosts: localhost

connection: local

become: yes

become_user: ansadm

tasks:

- name: Create .ssh directory

file:

path: /home/ansadm/.ssh

state: directory

owner: ansadm

group: ansadm

mode: '0700'

- name: Generate SSH key

openssh_keypair:

path: /home/ansadm/.ssh/id_rsa

type: rsa

size: 2048

owner: ansadm

group: ansadm

mode: '0600'

#####

PLAY 3: Copy SSH key to worker nodes

#####

- name: Configure passwordless SSH

hosts: workers

become: yes

tasks:

- name: Add controller public key to workers

authorized_key:

user: ansadm

key: "{{ lookup('file', '/home/ansadm/.ssh/id_rsa.pub') }}"

```
root@ip-172-31-13-122 playbooks]# cd
root@ip-172-31-13-122 ~]# cat ansible/hosts
controller
localhost ansible_connection=local

[workers]
worker-1 ansible_host=172.31.7.125
worker-2 ansible_host=172.31.15.0

[all:vars]
ansible_user=ec2-user
ansible_ssh_private_key_file=~/ssh/arj.pem
ansible_ssh_common_args=-o StrictHostKeyChecking=no

root@ip-172-31-13-122 ~]# cd /home/ec2
bash: cd: /home/ec2: No such file or directory
root@ip-172-31-13-122 ~]# cd playbooks/
root@ip-172-31-13-122 playbooks]# cat setup-passwordless.yml
---
#####
# PLAY 1: Create ansadm user on all servers
#####
- name: Create common user and sudo
  hosts: all
  become: yes

  tasks:
    - name: Create ansadm user
      user:
        name: ansadm
        shell: /bin/bash
        create_home: yes

    - name: Allow passwordless sudo
      copy:
        dest: /etc/sudoers.d/ansadm
        content: "ansadm ALL=(ALL) NOPASSWD:ALL"
        mode: '0440'

#####
# PLAY 2: Generate SSH key on controller (localhost)
#####
- name: Generate SSH key on controller
  hosts: localhost
  connection: local
  become: yes
  become_user: ansadm

  tasks:
    - name: Create .ssh directory
      file:
        path: /home/ansadm/.ssh
        state: directory
        owner: ansadm
```

→ ansible -I /root/ansible/hosts workers -m ping

```
root@ip-172-31-13-122 playbooks]# cd
root@ip-172-31-13-122 ~]# cd ansible/
root@ip-172-31-13-122 ansible]# vi hosts
root@ip-172-31-13-122 ansible]# ansible -i /root/ansible/hosts workers -m ping
[WARNING]: Platform linux on host worker-1 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
worker-1 | SUCCESS |
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
[WARNING]: Platform linux on host worker-2 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
worker-2 | SUCCESS |
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
```

→ ansible-playbook -i /root/ansible/hosts setup-passwordless.yml

```
ok: [worker-2]

TASK [Create ansadm user] *****
ok: [localhost]
changed: [worker-1]
changed: [worker-2]

TASK [Allow passwordless sudo] *****
ok: [localhost]
changed: [worker-1]
changed: [worker-2]

PLAY [Generate SSH key on controller] *****

TASK [Gathering Facts] *****
ok: [localhost]

TASK [Create .ssh directory] *****
ok: [localhost]

TASK [Generate SSH key] *****
ok: [localhost]

PLAY [Configure passwordless SSH] *****

TASK [Gathering Facts] *****
ok: [worker-1]
ok: [worker-2]

TASK [Add controller public key to workers] *****
changed: [worker-1]
changed: [worker-2]

PLAY RECAP *****
localhost                : ok=5  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
worker-1                  : ok=5  changed=3  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
worker-2                  : ok=5  changed=3  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0

[root@ip-172-31-13-122 playbooks]#
```

4. Ansible playbook to inject ansible vault variables.

```
root@ip-172-31-13-122:~/playbooks
db_user: admin
db_password: admin@123

[root@ip-172-31-13-122 playbooks]# vi vault.yml
[root@ip-172-31-13-122 playbooks]# cat vault.yml
db_user: admin
db_password: admin@123
[root@ip-172-31-13-122 playbooks]#
```

Add the data

db_user: admin

db_password: admin@123

From the same directory:

→ansible-vault encrypt vault.yml

You will be asked:

→New Vault password:

Confirm New Vault password:

After this, check the file:

cat vault.yml

Output will look like this:

```
[root@ip-172-31-13-122 playbooks]# ls
setup-passwordless.yml
[root@ip-172-31-13-122 playbooks]# vi vault.yml
[root@ip-172-31-13-122 playbooks]# cat vault.yml
db_user: admin
db_password: admin@123
[root@ip-172-31-13-122 playbooks]# ansible-vault encrypt vault.yml
bash: ansible-vault: command not found
[root@ip-172-31-13-122 playbooks]# mv vault.yml vault.yml
[root@ip-172-31-13-122 playbooks]# ansible-vault encrypt vault.yml
New Vault password:
Confirm New Vault password:
Encryption successful
[root@ip-172-31-13-122 playbooks]# cat vault.yml
$ANSIBLE_VAULT:1.1:AES256
62653937363430306663366437373432353066346130373937353462666236636634393932653537
3731386238643038386666343636643739333636343064383406666664353937361642862373561566
38366436663338343562666465363366562646666303433353838616330383733613830636632
3362623838336437330666326236534633665613666336131313034343832396461323030616366
666330376538643033393835383937396330333063343263313339653730346463633137396631
6133633331323166363639313938626138386166643839653063
[root@ip-172-31-13-122 playbooks]#
```

Since it's encrypted, you cannot view it with cat or vi directly anymore.

Here are the correct ways to view/edit vault data

Method 1:View vault contents (READ-ONLY)

→ansible-vault view vault.yml

It will ask:

Vault password:

After you enter the password, you'll see

```
[root@ip-172-31-13-122 playbooks]# ansible-vault view vault.yml
Vault password:
db_user: admin
db_password: admin@123
[root@ip-172-31-13-122 playbooks]#
```

db_user: admin

db_password: admin@123

Method 2: Edit vault contents (RECOMMENDED)

ansible-vault edit vault.yml

- Asks for vault password

- Opens file in editor
- Lets you modify values

Now we can able to edit

```
root@ip-172-31-13-122:~/playbooks
db_user: admin
db_password: admin@123

```

Let's you modify values

- Re-encrypts automatically on save

2) Create playbook to INJECT vault variables

—>  vault-playbook.yml

—> write below code

```
---
- name: Use Ansible Vault variables
  hosts: workers
  become: yes

  vars_files:
    - vault.yml

  tasks:
    - name: Print db user
      debug:
        msg: "DB User is {{ db_user }}"

    - name: Create file with secret
      copy:
        dest: /tmp/db.conf
        content: |
          username={{ db_user }}
          password={{ db_password }}
```

→ ansible-playbook -i hosts.ini vault-playbook.yml --ask-vault-pass

```
[root@ip-172-31-13-122 playbooks]# ansible-playbook -i /root/ansible/hosts vault-playbook.yml --ask-vault-pass
vault password:
PLAY [Use Ansible Vault variables] *****
TASK [Gathering Facts] *****
[WARNING]: Platform linux on host worker-1 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [worker-1]
[WARNING]: Platform linux on host worker-2 is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
ok: [worker-2]
TASK [Print db user] *****
ok: [worker-1] => {
  "msg": "DB User is admin"
}
ok: [worker-2] => {
  "msg": "DB User is admin"
}
TASK [Create File with secret] *****
changed: [worker-1]
changed: [worker-2]
PLAY RECAP *****
worker-1 : ok=3  changed=1  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
worker-2 : ok=3  changed=1  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
[root@ip-172-31-13-122 playbooks]#
```

ANSIBLE VAULT — ALL COMMANDS

Create encrypted vault file

→ ansible-vault create vault.yml

- Asks for vault password
- Creates encrypted file

✓ 2 **View encrypted file (read-only)**

→ansible-vault view vault.yml

- Asks vault password
- Shows content
- Does NOT allow editing

✓ 3 **Edit encrypted file (recommended)**

→ansible-vault edit vault.yml

- Asks vault password
- Opens file in editor
- Re-encrypts automatically on save

🔥 **Best practice**

✓ 4 **Encrypt existing file**

->ansible-vault encrypt vars.yml

- Encrypts normal YAML file

✓ 5 **Decrypt encrypted file**

→ansible-vault decrypt vault.yml

⚠ Makes file plain text

(Not recommended in production)

✓ 6 **Change vault password (key rotation)**

→ansible-vault rekey vault.yml

You will be asked:

Old vault password:

New vault password:

Confirm new vault password:

- ✅ Used when password expires or leaks
-

✅ 7 Run playbook with vault password (manual)

→ `ansible-playbook playbook.yml --ask-vault-pass`

Prompts for vault password during execution.

✅ 8 Run playbook using vault password file

Create file:

→ `vi ~/.vault_pass.txt`

Add password inside.

Set permission:

`chmod 600 ~/.vault_pass.txt`

Run:

`ansible-playbook playbook.yml \`

`--vault-password-file ~/.vault_pass.txt`

🔥 Used in Jenkins & CI/CD.

✅ 9 Encrypt only specific variables (inline vault)

`ansible-vault encrypt_string 'MyPassword123' --name 'db_password'`

Output example:

`db_password: !vault |`

`$ANSIBLE_VAULT;1.1;AES256`

`313233343536...`

Used inside vars files.

